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Unsupervised Machine Learning Workflow for Identifying Micro-Resistivity Borehole Image Features

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Abstract

The interpretation of micro-resistivity borehole image data is essential for geomechanical and geological analysis in the oil and gas industry but is often hindered by time-consuming manual processes. This study presents a novel unsupervised clustering workflow that leverages advanced deep learning models, including DINO for image feature extraction and LaMa for inpainting. The workflow combines several steps: borehole image inpainting, extracting features, normalizing, reducing dimensions, and clustering. Our findings show that this approach significantly improves the accuracy of borehole image restoration, as confirmed by metrics like RSNR, SSIM, and VSA. Furthermore, the workflow accurately identifies borehole features, such as breakouts and fractures validated by ARI, NMI, and F1 scores. The workflow also demonstrates strong adaptability across different borehole images. The significance of this research lies in its potential to enhance borehole image interpretation, offering a scalable and efficient solution that enhances operational decision-making and geomechanical model calibration.

Introduction

Geomechanical analysis for the oil and gas industry has experienced rapid growth over the past decades for both measurement and interpretation workflows. However, in the current state, such an analysis is still time-consuming and is often done in days or weeks after measurements. The majority of time is spent on gathering and processing multidisciplinary data from various measurements. Among all the input data, borehole image data, especially micro-resistivity data processing, is one of the key components.

Accurate interpretation of borehole images is vital. The reliance on manual interpretation methods is increasingly unsustainable due to the growing volume of data and the demand for faster, more reliable analysis. Furthermore, the inherent data gaps in borehole images, typically amounting to 40% of total coverage, pose significant challenges to geomechanical analyses. Addressing these issues through innovative machine learning approaches is crucial for enhancing operational efficiency and reducing costs.

This paper aims to introduce a novel unsupervised learning clustering workflow designed to overcome key challenges in the interpretation of micro-resistivity borehole images. Specifically, the workflow seeks to (1) effectively reconstruct missing areas within the images, (2) avoid the need for extensive, high-quality, and diverse training datasets by using an unsupervised approach, and (3) enhance generalization across

various borehole images from different geological basins. By providing an automated clustering solution, this study intends to streamline the interpretation process.

The primary contributions of this research are as follows:

1. **Development of a Comprehensive Workflow:** The paper presents an integrated unsupervised learning clustering workflow for borehole image interpretation, including inpainting, feature representation computation, normalization, dimensionality reduction, and clustering. This workflow systematically addresses the inherent challenges of borehole image analysis.
2. **Innovative Use of Deep Learning Models:** By leveraging state-of-the-art deep image representation models like the pretrained Dino models and LaMa inpainting model, the study introduces advanced techniques for reconstructing missing image areas and extracting robust feature representations, minimizing manual annotation efforts, and enhancing transfer learning capabilities.
3. **Benchmarking and Evaluation Metrics:** The research employs a variety of evaluation metrics, such as PSNR, SSIM, VSA, ARI, NMI and F1, to assess the quality of inpainting and clustering, ensuring rigorous validation of the proposed workflow.
4. **Field Case Study:** The paper includes a real-world application of the workflow in a field case study, demonstrating its practical utility in streamlining the calibration of mechanical earth models and aiding decision-making in field operations.

Through these contributions, the paper sets the stage for a transformative approach to borehole image interpretation, paving the way for improved scalability, efficiency, and accuracy in geomechanical and geological analyses.

Background and Related Work

The interpretation of borehole image data, particularly micro-resistivity imaging, emerges as a critical component in geomechanical and geological analysis. Traditional interpretation methods have relied heavily on manual annotation by experienced geologists, a process that is both time-consuming and subject to human bias and inconsistency. The inherent design limitations of borehole imaging tools result in considerable data gaps, with tool coverage typically reaching only 60%, leaving the remaining 40% as blank stripes that complicate accurate interpretation (Jiang, Si, and Wu 2024; Samarkin et al. 2023).

Supervised machine learning techniques have demonstrated significant potential in advancing borehole image interpretation in recent years. These approaches require extensive datasets of labeled training data, where geological features have been pre-identified by expert interpreters. Xu et al. (2025) demonstrate the effectiveness of neural network methods that integrate multi-layer semantic features for gap-filling in micro-resistivity imaging data. Also, supervised machine learning has been utilized for the combined interpretation of dipole sonic and borehole image data within geomechanical analysis (Keretchashvili et al. 2023). The study introduces a workflow which integrates cross-dipole dispersion analysis with micro-resistivity borehole image data to distinguish between different anisotropy mechanisms, including tool decentralization effects, breakouts, and layer-induced versus stress-induced azimuthal anisotropy.

Recognition of the limitations inherent in supervised approaches has driven the development of unsupervised methods for borehole image interpretation. These approaches address the fundamental challenge of obtaining sufficient labeled training data and the subjective nature of geological interpretation. Lima et al. (2019) proposes an approach which extracts representative information from images without relying on labeled data, using autoencoder reconstruction loss for network pre-training followed by joint training using cluster assignment hardening. Another significant contribution to unsupervised interpretation includes the work on Brazilian pre-salt ultrasonic borehole image data (Lima Angelo Dos Santos et al. 2022). The workflow integrates statistical methods and deep learning algorithms for image embedding, enabling data-driven and versatile borehole image interpretation through a five-step process that includes

data preparation, statistical embedding, dimensionality reduction using principal component analysis or t-SNE, clustering, and facies classification.

Self-supervised learning strategies also emerge as a promising alternative, particularly for addressing data gap issues. Jiang, Si, and Wu (2024) introduce a self-supervised approach that replicates gaps found in borehole images by randomly creating vertical blank strips to mask known areas, using the original images as labels to train networks for data recovery. Also, the paper introduced by Xu et al. (2025) proposes a two-step, deep-learning method to fill in missing sections of micro-resistivity imaging data, which first uses a conditional texture and structure dual generative (CTSDG) adversarial network to reconstruct the gaps, and then applies a diffusion model to enhance the image resolution. They use different evaluation metrics to evaluate the image reconstruction quality such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) which will also be used in the current study.

Overall, micro-resistivity borehole image interpretation has three fundamental constraints that hinder its scalability and generalization capabilities. First, inherent micro-resistivity imaging contains data gaps, which complicate accurate interpretation. Typically, these gap-filling methods are performed as a standalone step prior to further interpretation. Second, interpreting borehole images in a supervised manner requires extensive labeled datasets where geological features are manually annotated by domain experts – a process that is both time-intensive and subject to human bias. Third, supervised models sometimes exhibit poor transfer learning capabilities between geological basins due to domain shifts caused by feature distributions and imaging tool specifications.

Methodology

The primary aim of this paper is to introduce a robust unsupervised learning clustering workflow that tackles three significant challenges in the interpretation of micro-resistivity Borehole Images using machine learning techniques: (1) effectively reconstructing missing areas within the images, (2) reducing reliance on extensive, high-quality, diverse training datasets, and (3) addressing the lack of generalization in various borehole images from different geological basins. This innovative clustering workflow is designed to automatically cluster borehole image logs into regions characterized by distinct geological features. The workflow encompasses several key stages: inpainting, feature representation computation, normalization, dimensionality reduction, and clustering. Each step employs a variety of specialized techniques, which will be elaborated upon in subsequent sections of the paper.

Data Description

The dataset utilized in this study comprises fifteen real field micro-resistivity borehole images sourced from diverse geological environments. These high-resolution images capture the variations in electrical resistivity along the borehole walls, collected using advanced imaging tools. These images provide valuable insights into the geological and petrophysical characteristics of subsurface formations. These are 2D image datasets acquired using pad-based imagers with 360-degree coverage, capturing the resistivity variations along the borehole wall. The images are represented in an unwrapped cylindrical format, where the X-axis corresponds to the circumferential angle (0–360°), and the Y-axis represents the depth (ft), as shown in Fig. 1. The pixel values reflect variations in resistivity and are obtained using the dynamic version of micro-resistivity images, allowing for high-resolution visualization of formation features such as bedding, fractures, breakouts, and sedimentary structures. These imaging tools use a limited number of pads in contact with the borehole wall, the resulting images contain missing values in the gaps between pads, leading to incomplete circumferential coverage requiring additional processing techniques for full visual analysis.

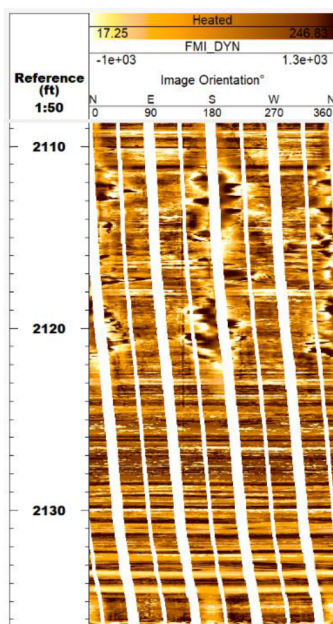


Figure 1—Sample of the micro-resistivity borehole image.

Unsupervised learning clustering workflow

Patch generation. Prior to implementing the clustering workflow, the raw borehole image log is split into smaller image patches, each with 360 pixels in depth and 360 pixels in orientation.

The clustering workflow is applied to this collection of patches, comprising multiple modules, each utilizing distinct algorithms. The workflow includes the following stages: inpainting, feature representation computation, normalization, dimensionality reduction, and clustering, as shown in Fig. 2.

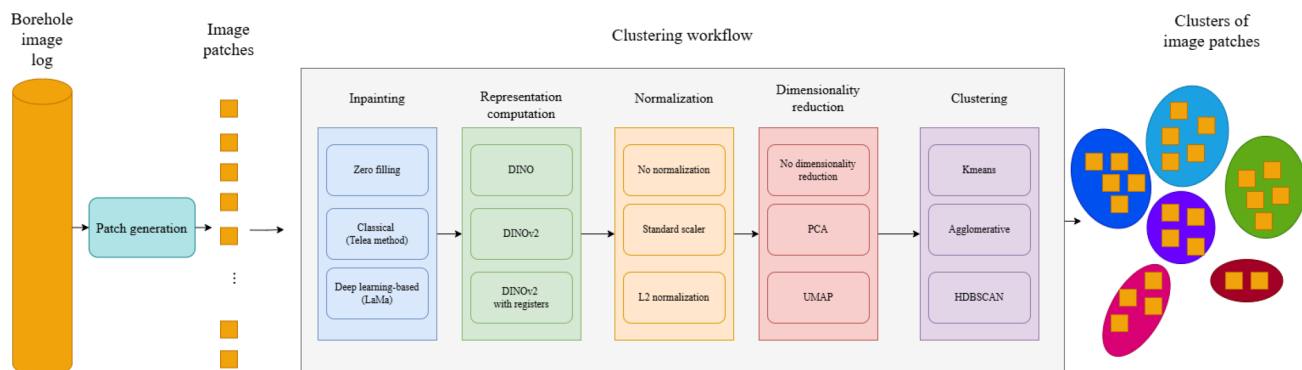


Figure 2—Unsupervised learning clustering workflow.

Inpainting. Inpainting is an integral component of the workflow due to the presence of missing pixels in micro-resistivity images, a common issue stemming from pad-based image acquisition. To reconstruct the complete image, various inpainting techniques can be employed. This workflow incorporates three distinct approaches: Zero filling, Classical inpainting, and Deep learning-based inpainting each with its own set of advantages and disadvantages.

- **Zero filling**

Zero filling involves substituting missing pixel values with zeros. This straightforward and widely used method provides a quick solution for addressing gaps in data and serves as an effective baseline for benchmarking evaluations.

- *Classical inpainting*

For the classical approach, we employed the Telea method, a well-established digital image restoration technique commonly implemented in OpenCV. This method reconstructs missing or damaged areas by propagating information from the surrounding known pixels. The algorithm works by using a fast-marching approach that estimates pixel values in the damaged area through weighted averaging of neighboring pixels, considering both the geometric distance and the smoothness of the underlying image structure. The method prioritizes filling pixels that are closer to the boundary of the missing region first, then progressively moves inward, ensuring that the reconstructed content maintains visual coherence with the surrounding areas (Canessa and Tenze 2023; Telea 2004). While Telea inpainting is computationally efficient, it may be outperformed by more advanced deep learning methods.

- *Deep learning-based inpainting*

Within the scope of deep learning inpainting, we have investigated the pretrained LaMa (Large Mask Inpainting) model proposed by Suvorov et al. (2021). The LaMa method is an advanced approach specifically designed to tackle the challenges of reconstructing images with extensive missing areas, intricate structures, and high resolutions. It introduces fast Fourier convolutions (FFCs) to achieve an image-wide receptive field, enabling the network to capture global context and periodic patterns effectively. The method combines three key components: A network architecture leveraging FFCs to manage long-range dependencies. A high receptive field perceptual loss that prioritizes structural coherence over pixel-level details and training with large randomly generated masks to improve performance on extensive damaged areas. LaMa's Fourier-based approach allows it to generalize to higher resolutions than those seen during training while maintaining computational efficiency.

These inpainting methods—Zero filling, Classical, and Deep learning—are employed directly for inference without additional training. Each method processes specific image patches as inputs and generates an inpainted image patch as a result. A comparative analysis of the inpainting quality will be presented in the results section.

Representation computation. Image feature extraction is a crucial step within the workflow, as it involves generating feature embeddings—compact, high-dimensional representations that encapsulate key visual characteristics of images, including color, texture, shape, and semantic content. These embeddings form the basis for numerous downstream tasks in computer vision. A well-crafted representation of image patches can effectively preserve significant information about the features within the images. However, given that borehole images differ significantly from natural images, leveraging pretrained models designed for natural images often proves challenging in extracting meaningful features from borehole images. However, the DINO (self-Distillation with NO labels) family of pretrained models represents a breakthrough in self-supervised learning for computer vision, with DINOv1 (Caron et al. 2021), DINOv2 (Oquab et al. 2023) and DINOv2 with registers (Darcet et al. 2023) offering progressively enhanced capabilities. DINOv1 pioneered a self-distillation framework where a student Vision Transformer (ViT) learns to predict the output distribution of a momentum-updated teacher network, leveraging multi-crop augmentation and contrastive learning without labels. This approach enabled ViTs to extract semantically rich features with explicit segmentation cues. DINOv2 scaled this paradigm by training a 1B-parameter ViT on a curated 142M-image dataset, combining automated data filtering with stability-focused optimizations to produce smaller, distilled models. The DINOv2 with registers variant introduced 4–6 additional learnable tokens to the input sequence, mitigating feature map artifacts in background regions and enabling smoother attention maps for dense prediction tasks.

The feature extraction robustness and model sizes are key differentiator among DINO family models (DINOv1, DINOv2, and DINOv2 with registers). Table 1 shows different dino models and their model sizes. The effectiveness of these methods in clustering tasks will be discussed in the results section.

Table 1—Names and Approximate Sizes of DINO Family Models

Model name	Model size (millions of parameters)
dino_vits8 / dino_vits16	~ 22
dino_resnet50	~ 24
dino_vitb8 / dino_vitb16	~ 86
dinov2_vits14 / dinov2_vits14_reg	~ 22
dinov2_vitb14 / dinov2_vitb14_reg	~ 87
dinov2_vitl14 / dinov2_vitl14_reg	~ 304
dinov2_vitg14 / dinov2_vitg14_reg	~ 1136

Normalization. Normalization is the process of adjusting the values in a dataset or vector (like a feature embedding) to follow a specific scale or distribution. In the context of feature embeddings extracted from images, normalization is commonly used as it helps standardize the feature space and makes distance-based methods more reliable. In this workflow, we offer three options: No Normalization, Standard Scaling, and L2 Normalization.

$$z = \frac{x - \mu}{\sigma} \quad \text{Eq. 1}$$

Where:

- x : The original value (or a data point) in the dataset.
- μ : The mean of the feature.
- σ : The standard deviation of the feature.
- z : The standardized value, also called the z-score.

$$\hat{x} = \frac{x}{\|x\|_2} = \frac{x}{\sqrt{x_1^2 + x_2^2 + \dots + x_n^2}} \quad \text{Eq. 2}$$

Where:

- $\hat{x}$: The L2-normalized version of vector x .
- x : The original vector.
- $\|x\|_2$: The L2 norm of the vector x , also called the Euclidean norm.

Dimensionality reduction. Dimensionality reduction is an optional module within the workflow, applied after the image feature computation and scaling stages. Dimensionality reduction after extracting pretrained feature embeddings serves to address the curse of dimensionality by compressing high-dimensional embeddings into lower-dimensional representations while preserving critical structural and semantic information (Kazempour et al. 2024; Zhang, Zhou, and Bollegala 2024). This process enhances computational efficiency, mitigates noise amplification, and facilitates downstream tasks such as clustering by reducing redundancy and isolating discriminative features.

This module incorporates two widely used dimensionality reduction techniques: [Principal Component Analysis \(PCA\) \(2002\)](#) and Uniform Manifold Approximation and Projection (UMAP) ([McInnes, Healy, and Melville 2018](#)).

PCA operates as a linear transformation that identifies orthogonal axes (principal components) maximizing variance in the data. It computes the covariance matrix of the input features, derives its eigenvectors and eigenvalues, and projects the data onto the top- k eigenvectors corresponding to the largest eigenvalues ([Daga et al. 2020](#); [Schlather and Reinbott 2021](#)).

UMAP employs a nonlinear strategy grounded in Riemannian geometry and algebraic topology. It constructs a weighted graph in high-dimensional space to model local neighborhoods and optimizes a low-dimensional embedding by preserving topological relationships through minimization of cross-entropy ([McInnes, Healy, and Melville 2018](#)). Unlike PCA, UMAP excels at retaining both local and global structures, making it superior for visualizing complex manifolds. For example, UMAP-generated embeddings of Fashion MNIST data reveal clearer cluster separation compared to PCA or t-SNE ([Lekschas 2023](#)). While PCA prioritizes variance, UMAP balances density-aware preservation and computational scalability, achieving faster runtimes than t-SNE ([Becht et al. 2019](#)).

Clustering. Clustering serves as the final module of the workflow, where input data is grouped into meaningful clusters in an unsupervised manner. This module features a selection of popular clustering algorithms, including k-means, agglomerative clustering, and HDBSCAN.

K-means clustering, introduced by MacQueen (1967) partitions data into k predefined clusters by iteratively minimizing within-cluster variance. It assigns points to the nearest centroid, recalculating centroids until convergence. Advantages include computational efficiency and scalability for large datasets, but it requires specifying k a priori, struggles with non-spherical clusters, and is sensitive to initial centroid placement ([Yang, Xie, and Yin 2023](#); [Xie and Fang 2019](#)).

Agglomerative hierarchical clustering merges clusters bottom-up based on linkage criteria (e.g., Ward's method) ([Murtagh and Legendre 2014](#); [Eric U. and Michael O. 2024](#)) producing dendrograms for flexible cluster extraction without predefining k . However, its $O(n^3)$ complexity limits scalability, and irreversible merge steps can propagate errors ([U. O. and O. O. 2024](#)).

HDBSCAN ([McInnes, Healy, and Astels 2017](#)) is an extension of DBSCAN ([Campello, Moulavi, and Sander 2013](#)), groups points based on density connectivity, automatically detects clusters of varying shapes/sizes and labeling outliers. It eliminates the need for k and handles noise robustly but struggles with high-dimensional data and requires careful tuning of density thresholds ([Rovida 2023](#); [Gray et al. 2023](#)).

The clustering methods are carefully chosen for this workflow due to their distinct advantages and limitations, ensuring a comprehensive approach to addressing the diverse characteristics of the image patch representations. K-means and Ward's method prioritize compact clusters but fail on complex geometries, while HDBSCAN addresses nonlinear patterns. K-means and HDBSCAN scale better than agglomerative methods. HDBSCAN's robustness to noise contrasts with k-means' sensitivity to outliers ([Gray et al. 2023](#); [Yang, Xie, and Yin 2023](#)).

Workflow validation and benchmarking procedure

In this workflow, we conducted numerical benchmarking to assess the quality of both inpainting and clustering, and we will outline the specific evaluation procedures applied to each task.

Inpainting task. To assess inpainting quality, we generated synthetic ground truth labels (masks) that provide the actual values of masked regions within image patches, enabling a direct comparison with predicted pixel values to accurately calculate reconstruction quality.

- *Synthetic ground truth mask generation*

To evaluate the quality of the different inpainting algorithms on the raw micro-resistivity images, we generate synthetic masks on each patch. As missing pixels in the image exhibit a vertical structure, we generated synthetic rectangular vertical masks for each image patch to mimic these missing regions. For evaluation purposes, we expanded these synthetic masks horizontally to assess not only the quality of reconstructed pixels within the masked area but also the transition between reconstructed and actual pixel values, thereby evaluating potential artifacts in the images. An example of both the synthetic mask and the evaluation mask (ground truth mask) is illustrated in Fig. 3.

- *Inpainting evaluation metrics*

To evaluate the quality of the reconstruction parts of the image patches, we utilize the following evaluation metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Variance Similarity Analysis (VSA), which are widely used metrics for evaluating image reconstruction quality. PSNR measures the ratio between the maximum possible pixel value and the power of distortion (error), providing a basic numerical assessment of reconstruction fidelity, with higher values indicating better quality. SSIM goes beyond pixel-wise comparison by assessing structural similarity, considering luminance, contrast, and texture, making it more aligned with human visual perception. VSA evaluates the similarity in pixel-level variance patterns between the original and reconstructed images, capturing how well local variations and texture details are preserved—complementing PSNR and SSIM with a focus on statistical consistency.

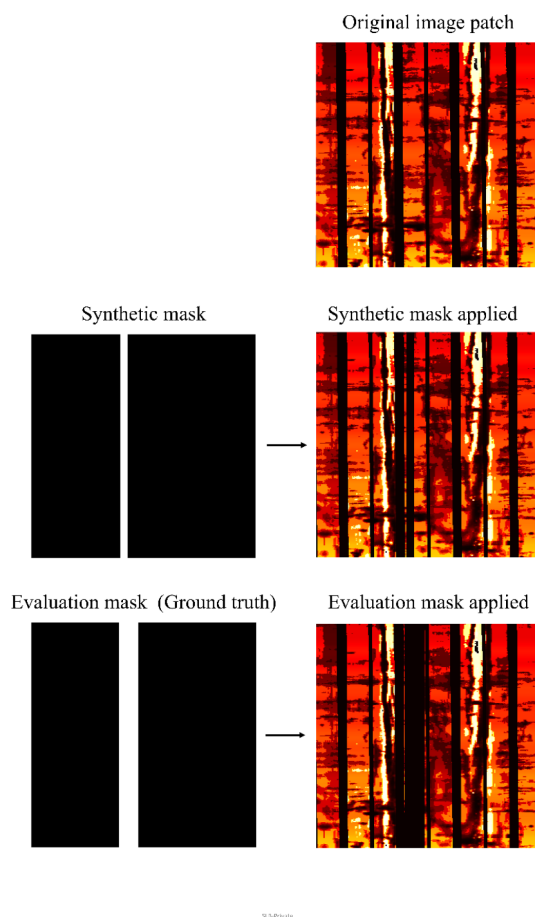


Figure 3—Example of synthetic and ground truth masks on a sample image patch.

PSNR:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad \text{Eq. 3}$$

Where:

- $\text{MSE} = \frac{1}{N} \sum_{i=1}^N (I_i - \hat{I}_i)^2$
- (MAX): Maximum possible pixel value
- (N): Number of pixels in the masked region
- (I_i): Original pixel value.
- ($\hat{I}_i$): Reconstructed pixel value.

SSIM:

$$\text{SSIM}(I, \hat{I}) = \frac{(2\mu_I \mu_{\hat{I}} + C_1)(2\sigma_{I\hat{I}} + C_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + C_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + C_2)} \quad \text{Eq. 4}$$

Where:

- (μ_I): Mean intensity of the original image.
- ($\mu_{\hat{I}}$): Mean intensity of the reconstructed image.
- (σ_I^2): Variance of the original image.
- ($\sigma_{\hat{I}}^2$): Variance of the reconstructed image.
- ($\sigma_{I\hat{I}}$): Covariance between the original and reconstructed images.
- (C_1), (C_2): Small constants to stabilize the division.

VSA:

$$\text{VSA} = 1 - \frac{|\sigma_I^2 - \sigma_{\hat{I}}^2|}{\sigma_I^2 + \sigma_{\hat{I}}^2 + \epsilon} \quad \text{Eq. 5}$$

Where:

- (σ_I^2): Variance of the original image in the masked region.
- ($\sigma_{\hat{I}}^2$): Variance of the reconstructed image in the masked region.
- (ϵ): Small constant to avoid division by zero.

Clustering task.

- *Ground truth generation*

For numerical evaluation purposes in the clustering task, we manually labeled image patches into three distinct classes: "breakout," "noise," and "other" - which encompass the remaining patches.

- *Clustering evaluation metrics*

To evaluate clustering quality, we utilize Adjusted Rand Index (ARI), which computes a similarity measure between two clusterings by considering all pairs of samples and counting pairs that are assigned in the same or different clusters in the predicted and true clustering. Normalized Mutual Information (NMI) is a normalization of the Mutual Information (MI) score to scale the results between 0 (no mutual information) and 1 (perfect correlation) and the F1 score comparing the clusters found by an algorithm against a ground truth classification. It combines precision and recall into a single metric to provide a balanced measure of accuracy, to specifically assess the accuracy and effectiveness of clustering these groups.

ARI:

$$ARI = \frac{\sum_{ij} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] / \binom{n}{2}} \quad \text{Eq. 6}$$

Where:

- (n_{ij}) is the number of objects in both cluster (i) of the ground truth and cluster (j) of the predicted clustering.
- (a_i) and (b_j) are the sums over rows and columns of the contingency table (n) is the total number of samples.

NMI:

$$NMI(U, V) = \frac{2 \times I(U; V)}{H(U) + H(V)} \quad \text{Eq. 7}$$

Where:

- $I(U; V)$ is the Mutual Information between clusterings U and V:

$$\begin{aligned} I(U; V) &= \sum_{i=1}^R \sum_{j=1}^C p_{ij} \log \frac{p_{ij}}{p_i p_j} \\ H(U) &= - \sum_{i=1}^R p_i \log p_i \\ H(V) &= - \sum_{j=1}^C p_j \log p_j \end{aligned} \quad \text{Eq. 8}$$

F1 score:

$$\begin{aligned} F_1 &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ \text{Precision} &= \frac{TP}{TP + FP} \\ \text{Recall} &= \frac{TP}{TP + FN} \end{aligned} \quad \text{Eq. 9}$$

Where:

- TP = Number of pairs of points that are in the same cluster in both predicted and true labels (True Positives)
- FP = Number of pairs in the same cluster in predicted labels but different in true labels (False Positives)

- FN = Number of pairs in the same cluster in true labels but different in predicted labels (False Negatives)

In the evaluation section, we aim to ensure a fair comparison among the clustering methods: k-means, agglomerative clustering, and spectral clustering. To achieve this, we will generate the same predefined number of clusters for k-means and agglomerative clustering across these methods and conduct multiple runs to assess stability and consistency in the results. While we will also include HDBSCAN in our evaluation, it differs from the other methods in that it does not require a predefined number of clusters, making direct comparison challenging. Instead, HDBSCAN necessitates careful hyperparameter tuning to determine the optimal clustering configuration, which may affect comparability with methods that rely on a fixed cluster count.

Results and Discussion

This section presents and discusses the results derived from the proposed workflow for inpainting and clustering tasks. The evaluation was conducted using fifteen distinct borehole micro-resistivity datasets, ensuring a comprehensive assessment of the workflow's capabilities.

Inpainting task

For the inpainting tasks, performance metrics such as PSNR, SSIM, and VSA were employed, as outlined in the methodology section. [Table 2](#) provides a concise summary of the evaluation outcomes across various inpainting techniques, including zero-filling, classical approaches, and deep learning models, applied to all the image patches on the masked region only. [Table 3](#) provides a concise summary of the same results on patches containing breakouts and fractures, to clearly see the difference on reconstruction quality on the image patches containing target classes.

Table 2—Benchmarking results on all the image patches across all the borehole images.

Model	PSNR	SSIM	VSA
Zero filling	8.77 ± 1.93	0.47± 0.18	0.70± 0.15
Classical	23.90± 3.91	0.81± 0.11	0.95± 0.04
Deep learning	24.19± 4.23	0.83± 0.10	0.99± 0.02

Table 3—Benchmarking results on the image patches containing breakouts and fractures across all the borehole images.

Model	PSNR	SSIM	VSA
Zero filling	9.55 ± 1.81	0.51± 0.16	0.73± 0.16
Classical	24.75± 3.41	0.85± 0.09	0.97± 0.02
Deep learning	25.79± 3.57	0.87± 0.08	0.99± 0.01

The results show that deep learning outperforms classical inpainting in all evaluation metrics. The sample inpainted visualization ([Fig. 4](#)) clearly supports the numerical results. The deep learning inpainting algorithm excels in reconstruction, effectively minimizing vertical artifacts while preserving the structural integrity of objects within the image.

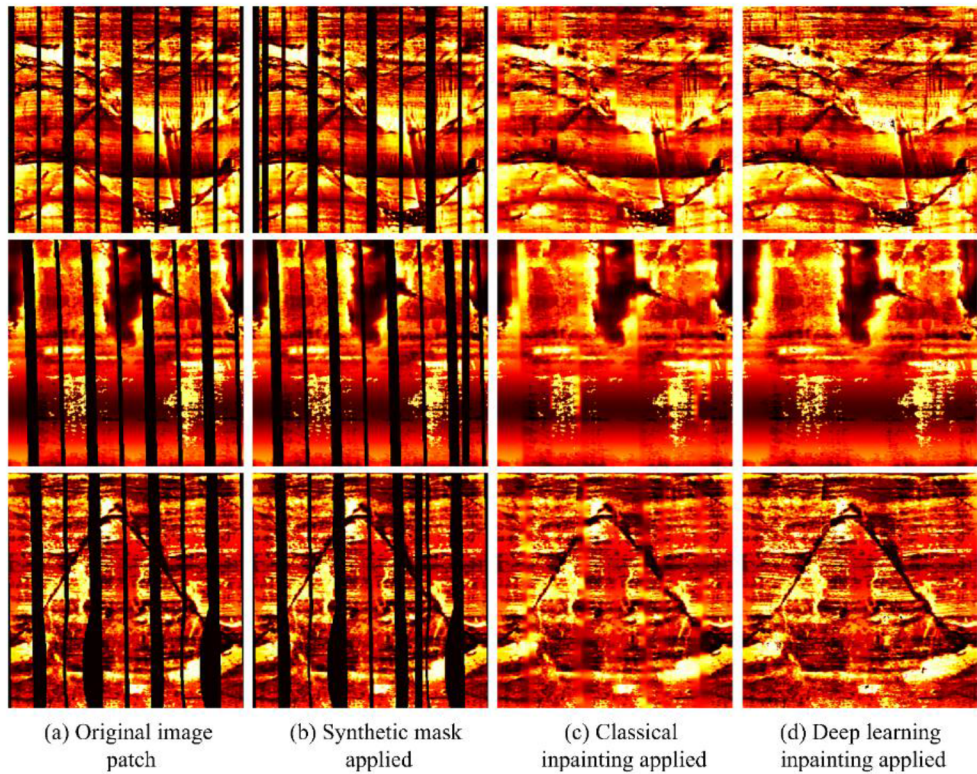


Figure 4—Comparison of inpainting algorithms: (a) Original image patch, (b) Synthetic mask applied, (c) Classical inpainting algorithm applied, (d) Deep learning inpainting algorithm applied.

Evaluation of Clustering Workflow Components: An Ablation Study

An ablation study is a systematic method used to evaluate the impact of individual components within a workflow on their overall performance. In the context of clustering tasks, metrics such as ARI, NMI, and average F1 score are employed, alongside an aggregate metric that averages ARI, NMI, and F1 scores considering both breakout and noise clusters. [Table 4](#) highlights the statistics of the two highest-performing combinations across all clustering algorithms. The results indicate that HDBSCAN achieved the highest aggregate score, followed by agglomerative clustering and K-means.

Table 4—Benchmarking results of the top two best combinations of each clustering algorithm averaged across all datasets.

Inpainting	Representation	Normalization	Dim reduction	Clustering	F1 average	ARI (partial)	NMI (partial)	Aggregate
classical	dino_vitb16	Standard	None	hdbscan	0.71 ±0.0	0.98 ± 0.0	0.94± 0.0	0.88±0.0
classical	dino_vitb8	Standard	None	hdbscan	0.7 ±0.0	0.98 ± 0.0	0.94± 0.0	0.88±0.0
deep_learning	dinov2_vitb14_reg	L2	None	agglomerative	0.8 ±0.0	0.7 ± 0.0	0.72± 0.0	0.74±0.0
fill_with_zero	dino_vitb16	Standard	UMAP	agglomerative	0.87 ±0.02	0.66±0.06	0.7±0.04	0.74±0.04
classical	dinov2_vitg14_reg	Standard	UMAP	kmeans	0.86 ±0.0	0.63±0.01	0.66±0.01	0.72±0.01
classical	dinov2_vitl14	Standard	UMAP	kmeans	0.86 ±0.02	0.62±0.02	0.66±0.02	0.72±0.0

To determine the contribution of each module to clustering performance, we performed one-way ANOVA analysis ([Lakens 2013](#)) for each module-metric combination. Effect sizes are calculated using eta-squared (η^2), which represents the proportion of variance in each metric explained by each module. Effect sizes are interpreted as negligible ($\eta^2 < 0.01$), small (0.01-0.06), medium (0.06-0.14), or large ($\eta^2 > 0.14$).

In Fig. 5 effect size (η^2) heatmap (a) provides a visual representation of the contribution strength of each module across different metrics (ARI, NMI, F1 avg). Modules with higher η^2 values indicate a stronger influence on the corresponding metric. The color gradient highlights the magnitude of the effect size, with darker shades representing larger contributions. This visualization helps identify which modules have the most significant impact on clustering performance. Whereas bar chart (b) in the same figure highlights the overall module importance and ranks modules based on their average effect size (η^2) across all metrics. Modules are categorized into four levels of contribution: negligible, small, medium, and large, as indicated by the color-coded of the bars. Horizontal dashed lines mark the thresholds for these categories.

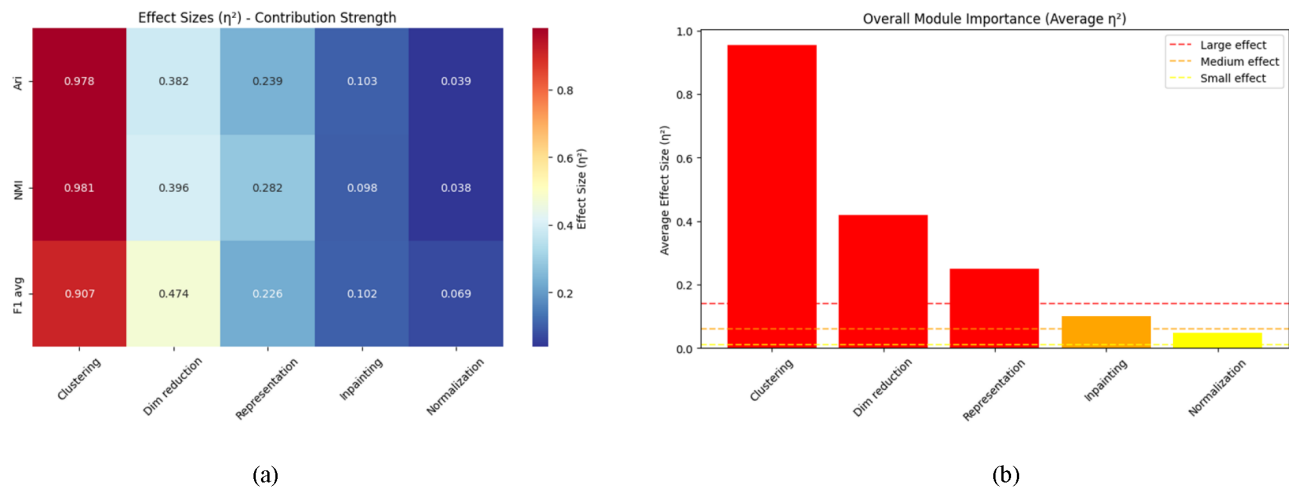


Figure 5—The effect size heatmap (a) showing the influence of each module on clustering metrics, with darker colors indicating stronger contributions. The bar chart (b) ranks modules by their average effect size, highlighting key contributors to each module.

The bar chart clearly illustrates that choosing the appropriate clustering algorithm, employing effective dimension reduction techniques, and utilizing robust image representation are crucial for clustering borehole image features optimally. In contrast, inpainting has a moderate impact, while normalization plays a minor role, possibly because the embedding of the DINO family is inherently normalized.

Case Study

This section presents a field case study to demonstrate the workflow for automatically inpainting and clustering micro-resistivity borehole image data (Donald et al. 2021). Additionally, to illustrate how this workflow can benefit decision-making in real-time field applications, we show how the labeled clusters of borehole breakouts can enable a streamlined calibration of the Mechanical Earth Model.

The acquired borehole image and inpainted borehole image are plotted in tracks 2 and 3 of Fig. 6, respectively. Track 4 displays the clustering labels in distinct colors. In total, thirty-eight clusters are identified, representing several types of micro-resistivity image features caused by formation heterogeneity, which indicated by the gamma-ray log in track 1. Among all the clusters, cluster label number 7, separately plotted in track 5, is found to correlate well with breakouts in the micro-resistivity image. Cluster number 28, also plotted in track 5, is well correlated with drilling induced fractures or high dipping angle fractures. To better illustrate this observation, we selected six zones that clearly show the correlation, as depicted in Fig. 7 (a) to (f), respectively.

Finally, we apply the clustering results to calibrate a Mechanical Earth Model (MEM) (Plumb et al. 2000). The MEM is an important framework in the geomechanical workflow. It mainly involves the construction of the rock mechanical properties, pore pressure, and in-situ principal stresses to help predict and avoid drilling risk, design a hydraulic fracture plan, and optimize the perforation and production.

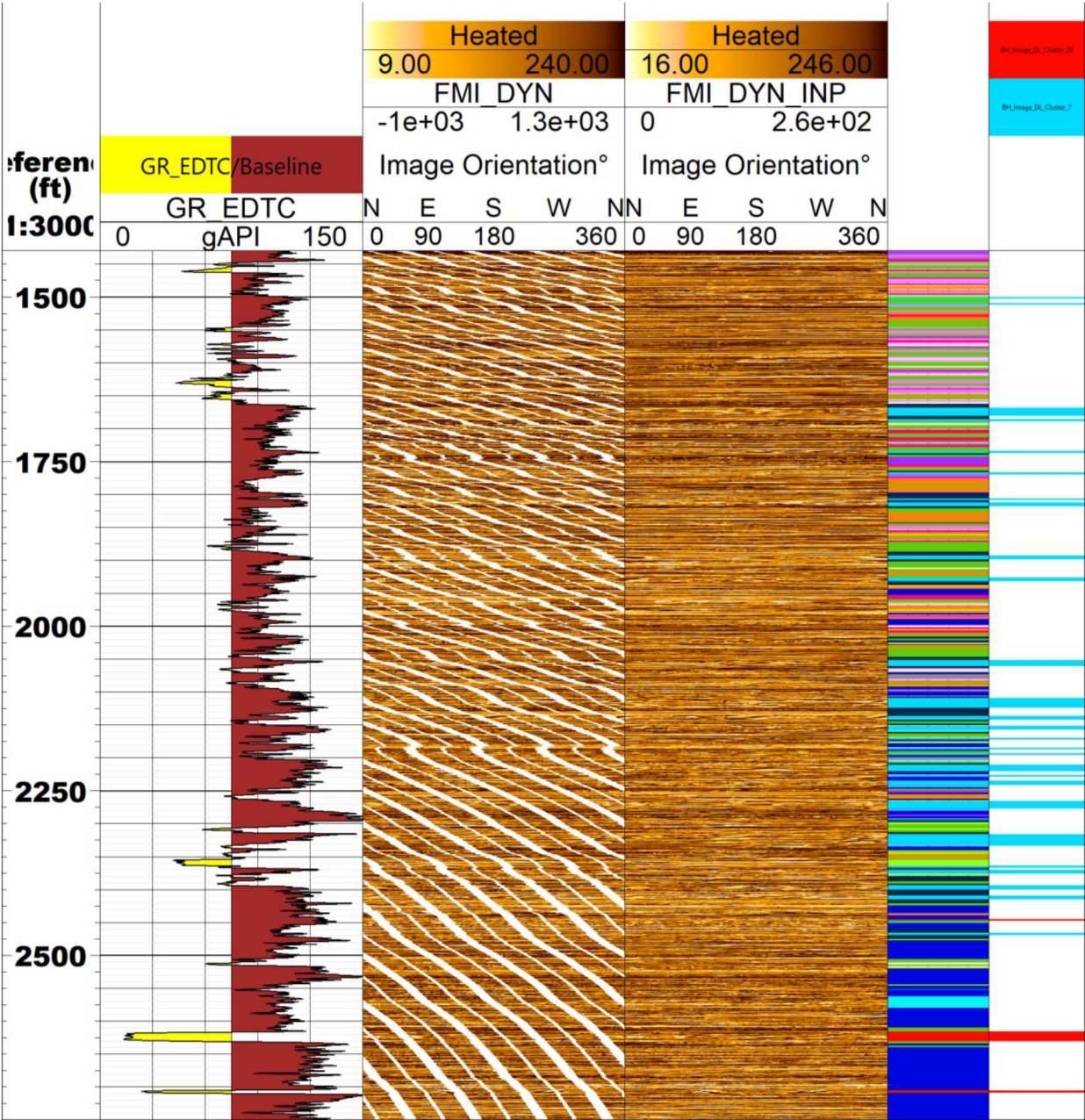


Figure 6—The field case study of inpainting and clustering results using the developed unsupervised machine learning workflow. Track 1: gamma-ray; Tracks 2 and 3: Acquired and inpainted micro-resistivity images; Track 4: Clustering labels in distinct colors; Track 5: Clustering labels that correspond to borehole breakout and fracture features.

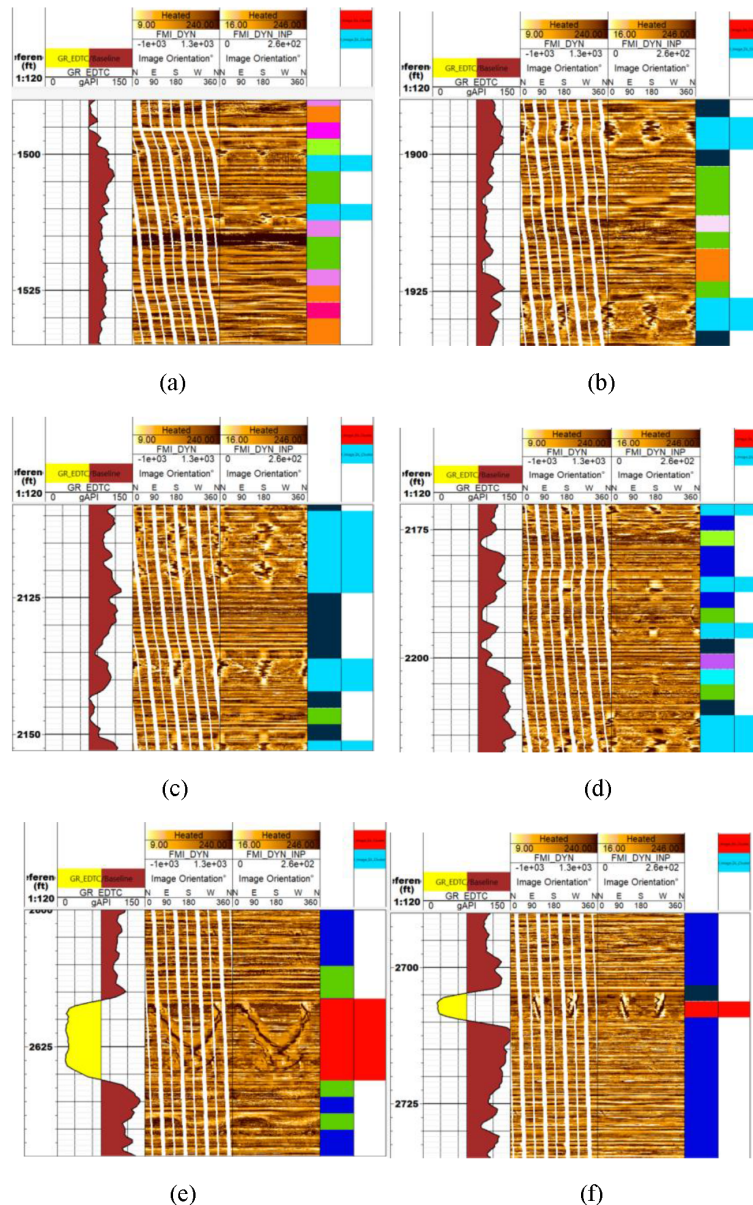


Figure 7—Six zones (a)-(f) that are selected from the field case study to highlight the correlation between clustering labels with both breakouts and fractures observed from the micro-resistivity image.

The rock mechanical properties are derived from borehole sonic data, which provide dynamic elastic moduli. Static elastic moduli and rock strength can be obtained through a core-based correlation method. Overburden stress is computed by integrating density log over depth. The in-situ horizontal stresses, particularly the maximum horizontal stress, are not directly measured and are often estimated using the poro-elastic strain model (Thiercelin and Plumb 1994). Two calibration parameters, E_{xx} and E_{yy} , are tuning parameters to account for the tectonic strains and other effects. These parameters are often calibrated based on rock shear failure (Birchwood et al. 2021). This calibration process is often time-consuming due to the need to label breakouts and fractures from borehole images manually. With the labeling results obtained from the automatic machine learning workflow, this calibration process can be significantly streamlined.

Fig. 8 shows the calibrated MEM of the field case study using the breakout labels detected by the machine learning workflow. Track 1 plots the borehole deviation and azimuth. Track 2 plots the rock mechanical properties, including static Young's modulus (YME_STA), static Poisson's ratio (PR_STA), Biot constant (ALPH_STA), tensile strength (TSTR), and uniaxial compressive strength (UCS), derived from borehole

		Mechanical properties		Pressures		Shear failure (3degree - North)			
		YME_STA 0 Mpsi 20							
		ALPH_STA unitless 1							
		PR_STA unitless 0.5		PPRS_LOG psi 15000		ShallowKnockout 0.00 1.00			
		UCS psi 20000		TZSP_LOG psi 15000		BF_SSKO_IMAGE (deg) 360			
		TSTR psi 20000		SHMAX_PHS psi 15000		WideBreakout 0.00 1.00		BH_Image_DL_Cluster_7	
		FANG deg 60		SHMIN_PHS psi 15000		BF_SWBO_IMAGE (deg) 360		BH_Image_DL_Cluster_7	
HAZIM deg 360		DEVIM deg 90							
TYD (ft) 1:1000	0	0	0	0	0	0	0	0	0
	1500								
	1600								
	1700								
	1800								
	1900								
	2000								
	2100								
	2200								
	2300								
	2400								

Conclusion

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demonstrate that the proposed methodology improves accuracy in feature detection and offers robust generalization across various borehole image environments. This solution significantly impacts users by streamlining the borehole image interpretation process, reducing manual effort, and enhancing decision-making capabilities in geomechanical and geological applications. By automating key interpretation tasks, the workflow provides a scalable and efficient tool for the oil and gas industry, facilitating timely and informed operational decisions.

Future efforts will focus on validating the workflow with diverse geological datasets to ensure robustness and adaptability. We also plan to assess its performance on different types of borehole images to evaluate cross-domain applicability. Incorporating few-shot learning techniques and human-in-the-loop elements could enhance the workflow by enabling learning from limited data while integrating expert feedback. Additionally, the inclusion of sonic measurements as inputs could enrich the interpretation process, offering a more comprehensive analysis. Finally, hybrid models that combine unsupervised and semi-supervised approaches will be investigated to further refine predictive capabilities, paving the way for more adaptive and user-centric solutions.

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